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* Fisher-Cochran Theorem

Sup that $Y \sim N_n(\mu, I)$ and $A_1, \dots, A_r$ $n \times n$ symmetric positive semidefinite matrix s.t. $Y^T Y = \sum_{j=1}^r Y^T A_j Y$ -- (*)

Then $Y^T A_j Y \sim \chi^2_{P(A_j)} (\lambda_j/2)$ for all $j=1, \dots, r$ and these quadratic forms are independent if and only if

$\sum_{j=1}^r P(A_j) = n$ in which case $\lambda_j = \mu^T A_j \mu$, $j=1, \dots, r$ & $\sum_{j=1}^r \lambda_j = \mu^T \mu$

Application: Determine distribution of individual components in an ANOVA decomposition.

Proof: Only if: $Y^T A_j Y \sim \chi^2_{P(A_j)} (\lambda_j/2)$ and their independence implies that $\exists$ random vectors $X_1, \dots, X_r$ s.t.

$$X_j \underset{P(A_j|X)}{\overset{\text{indep}}{\sim}} N_{P(A_j)}(\theta_j, I)$$

s.t. $Y^T A_j Y \stackrel{d}{=} X_j^T X_j$ in particular $\lambda_j = \|\theta_j\|^2$ for $j=1, \dots, r$
Want to show that $\sum_{j=1}^r P(A_j) = n$

By (*) $Y^T Y = \sum_{j=1}^r Y^T A_j Y \stackrel{d}{=} \sum_{j=1}^r X_j^T X_j = \|\tilde{X}\|^2$ ($\tilde{X} = (X_1, \dots, X_r)^T$)

Also $Y^T Y \sim \chi^2_{(n)} (\frac{\|\mu\|^2}{2})$

Let $k = \sum_{j=1}^r P(A_j)$. $\tilde{X} \sim N_k \left(\begin{pmatrix} \theta_1 \\ \vdots \\ \theta_r \end{pmatrix}, I \right)$

$\therefore \|\tilde{X}\|^2 \sim \chi^2_k \left(\frac{\sum_{j=1}^r \|\theta_j\|^2}{2} \right)$
 $= \chi^2_k \left(\frac{\sum_{j=1}^r \lambda_j}{2} \right)$

$Y^T Y \stackrel{d}{=} \|\tilde{X}\|^2$

We have $k = n$, $\sum_{j=1}^r \lambda_j = \|\mu\|^2 = \mu^T \mu$

$E\|X_j\|^2 = \lambda_j + P(A_j)$ -- (i)

$$\begin{aligned}\Rightarrow E(Y^T A_j Y) &= E(\text{tr}(A_j Y Y^T)) \\ &= \text{tr}(A_j E(Y Y^T)) \\ &= \text{tr}(A_j (\mu \mu^T + I)) \\ &= \mu^T A_j \mu + \text{tr}(A_j)\end{aligned}$$

$$\Rightarrow \lambda_j + P(A_j) = \mu^T A_j \mu + \text{tr}(A_j) \quad \text{-- (ii)}$$

Enough to show A_j are idempotent.

Conversely, s.p.s. $\sum_{j=1}^r P(A_j) = n$, start w/ rank-factorizⁿ of A_j 's. There exist $C_j^{j=1, \dots, r}$ that're of full col-rank s.t.

$$A_j = C_j C_j^T$$

$$\text{By } \star Y^T Y = \sum_{j=1}^r Y^T A_j Y = Y^T \sum_{j=1}^r C_j C_j^T Y = Y^T B B^T Y \text{ where}$$

$B = [C_1 \dots C_r]$. By our assumption, B is an $n \times n$ matrix. $Y^T B B^T Y = Y^T Y$ for $Y \sim N(\mu, I)$ $\forall Y \in \mathbb{R}^n$. By spectral decomposition of $B B^T$, all the eigenvalues of $B B^T$ must be equal to 1. $\Rightarrow B B^T = I_n$

$$\Rightarrow B^T B = I_n \quad \text{-- (iii)}$$

$\Rightarrow B$ is orthogonal.

$$\text{By (iii), } I_n = \begin{bmatrix} C_1^T C_1 & C_1^T C_2 & \dots & C_1^T C_r \\ & C_2^T C_2 & \dots & C_2^T C_r \\ & & \ddots & \\ & & & C_r^T C_r \end{bmatrix}$$

$$\Rightarrow C_j^T C_j = I_{P(A_j)} \text{ \& } C_j^T C_{j'} = 0 \quad \forall j \neq j'$$

$\Rightarrow$ columns of C_j are orthonormal.

$$\begin{aligned}\text{So the other have } C_j C_j^T &= A_j \text{ so that } A_j A_j = C_j C_j^T C_j C_j^T \\ &= C_j I_{P(A_j)} C_j^T \\ &= C_j C_j^T = A_j\end{aligned}$$

$$\Rightarrow A_j^2 = A_j$$

→ Inference under Gauss Markov setup

Ex: $Y = \beta_0 + \beta_1 X^{(1)} + \dots + \beta_{p-1} X^{(p-1)} + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I)$

$$H_0: \begin{pmatrix} \beta_1 = \beta_2 \\ \beta_2 + \beta_3 = 0 \end{pmatrix}$$

$$\beta_0 + \beta_1 X^{(1)} + \beta_2 X^{(2)} + \beta_3 X^{(3)} \quad (p=4)$$

under H_0 $\beta_0 + \beta_1 (X^{(1)} + X^{(2)} - X^{(3)})$

If it was rather $H_0: \begin{pmatrix} \beta_1 = \beta_2 \\ \beta_2 + \beta_3 = 1 \end{pmatrix}$, then:

$$\beta_0 + \beta_1 X^{(1)} + \beta_2 X^{(2)} + \beta_3 X^{(3)} \xrightarrow{H_0} \beta_0 + \beta_1 (X^{(1)} + X^{(2)} - X^{(3)}) + X^{(3)}$$

- General form of the null hypothesis specifying a reduced model (a submodel) is

$$H_0: \underset{r \times p \quad p \times 1}{L^T} \underset{r \times 1}{\beta} = \underset{r \times 1}{\theta_0}$$

- Assumptions: (i) $P(X) = p$
(ii) $P(L) = r \leq p$

We want to obtain the LS estimates of the parameter β under the constraint $\theta = \theta_0$ where $\theta = L^T \beta$

- Constraint Least Squares (CLS) is the following optimization problem:

$$\min_{\beta} \|Y - X\beta\|_2^2 \text{ subject to } L^T \beta = \theta_0 \quad \text{--- } \textcircled{\star} \text{ CLS}$$

Lagrange multiplier method:

$$\mathcal{L}(\beta, \lambda) = \frac{1}{2} \|Y - X\beta\|^2 + \lambda^T (L^T \beta - \theta_0)$$

$$\frac{\partial \mathcal{L}}{\partial \beta} = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 0$$

Transformation: $X_0 = (X^T X)^{-1/2} X$

$$Y = X\beta + \varepsilon, \quad X_0^T X_0 = I_p$$

$$= X_0 \gamma + \varepsilon$$

$$X\beta = X(X^T X)^{-1/2} (X^T X)^{1/2} \beta = X_0 \gamma$$

$$\gamma = (X^T X)^{1/2} \beta$$

$$\theta_0 = L^T \beta = L_0^T \gamma \text{ where } L_0 = \underset{p \times p}{(X^T X)^{-1/2}} \underset{p \times r}{L}$$

* (A) CLS is equiv. to

$$\min_{\gamma} \frac{1}{2} \|Y - X_0 \gamma\|_2^2 + \lambda^T (L_0^T \gamma - \theta_0)$$

$$\mathcal{L}(\gamma, \lambda)$$

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial \gamma} = -X_0^T (Y - X_0 \gamma) + L_0 \lambda = 0$$

$$\Rightarrow \gamma + L_0 \lambda = X_0^T Y \quad \text{-- (i)}$$

$$\text{also } L_0^T \gamma = \theta_0 \quad \text{-- (ii)}$$

$$\text{Multiply (i) by } L_0^T \Rightarrow L_0^T \gamma + L_0^T L_0 \lambda = L_0^T X_0^T Y$$

$$\Rightarrow L_0^T L_0 \lambda = L_0^T X_0^T Y - \theta_0$$

$$\Rightarrow \lambda = (L_0^T L_0)^{-1} (L_0^T X_0^T Y - \theta_0)$$

Plugging λ directly into (i) we get:

$$\tilde{\gamma} = X_0^T Y - L_0 (L_0^T L_0)^{-1} (L_0^T X_0^T Y - \theta_0)$$

Notice that the unconstrained LSE of γ is

$$\hat{\gamma} = X_0^T Y \because X_0^T X_0 = I_p$$

$$\dots \tilde{\gamma} = \hat{\gamma} - L_0 (L_0^T L_0)^{-1} (L_0^T X_0^T Y - \theta_0)$$

$$= \hat{\gamma} - L_0 (L_0^T L_0)^{-1} (L_0^T \hat{\gamma} - \theta_0)$$

$$= \hat{\gamma} - L_0 (L_0^T L_0)^{-1} (\hat{\theta} - \theta_0) \rightarrow \text{BLUE for } \theta$$

Propⁿ:

Under the assumption $p(X) = p \geq r = p(L)$ the ~~least~~ LSE of β under the constraint $L^T \beta = \theta_0$ is given by:

$$\tilde{\beta} = \hat{\beta} - (X^T X)^{-1} L (L^T (X^T X)^{-1} L)^{-1} (\hat{\theta} - \theta_0) \text{ where } \hat{\theta} = L^T \hat{\beta} \text{ \&}$$

$\hat{\beta} = (X^T X)^{-1} X^T Y$ is the unconstrained LSE of β .

→ Reduction in SSE when we go from the constrained to the unconstrained model

$$\|Y - X\tilde{\beta}\|^2 - \|Y - X\hat{\beta}\|^2$$

$\hat{y}^2 \leftarrow$ fitted under constrained model

$$\hookrightarrow = ||Y - P_X Y||^2 + ||P_X Y - X \tilde{\beta}||^2$$

$$= \|Y - X\hat{\beta}\|^2 + \|X\hat{\beta} - X\tilde{\beta}\|^2$$

By using the propⁿ:

$$\|X(\hat{\beta}^0 - \tilde{\beta})\|^2 = (\hat{\theta} - \theta_0)^T (L^T (X^T X)^{-1} L)^{-1} (\hat{\theta} - \theta_0)$$

Notice that $\text{var}(\hat{\theta}) = \text{Var}(L^T \hat{\beta}) = L^T \text{Var}(\hat{\beta}) L = \sigma^2 (L^T (X^T X)^{-1} L)$